CP1.9: Vectors in 3D

1. 3D Vector Basics

A 3D vector is written as $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

Magnitude: $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
Unit vector: $\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{|\mathbf{a}|}$

Direction Cosines

The direction cosines of a vector $\mathbf{a} = (a_1, a_2, a_3)$ are:

$$\cos\alpha = \frac{a_1}{|\mathbf{a}|}, \quad \cos\beta = \frac{a_2}{|\mathbf{a}|}, \quad \cos\gamma = \frac{a_3}{|\mathbf{a}|}, \quad \text{where } \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

2. Scalar (Dot) Product

Geometric definition:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$

where θ is the angle between the vectors $(0 \leq \theta \leq \pi)$.

Component form:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Key Applications

- **Testing perpendicularity:** $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$
- **Finding angle:** $\cos\theta = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$
- **Projection of $\mathbf{a}$ onto $\mathbf{b}$:** $\dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$

Properties

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ (commutative)
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ (distributive)
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

3. Cross (Vector) Product

Geometric definition:

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin\theta \, \hat{\mathbf{n}}$$

where θ is the angle between them and $\hat{\mathbf{n}}$ is the unit vector perpendicular to both, following the right-hand rule.

Component form (determinant expansion):

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \mathbf{i}(a_2 b_3 - a_3 b_2) - \mathbf{j}(a_1 b_3 - a_3 b_1) + \mathbf{k}(a_1 b_2 - a_2 b_1)$$

Key Properties of Cross Product

- $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ (anti-commutative)

- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$
- $\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} \times \mathbf{b} = \mathbf{0}$
- $\mathbf{i} \times \mathbf{j} = \mathbf{k}, \mathbf{j} \times \mathbf{k} = \mathbf{i}, \mathbf{k} \times \mathbf{i} = \mathbf{j}$

Area Applications

$$\text{Area of parallelogram} = |\mathbf{a} \times \mathbf{b}|$$

$$\text{Area of triangle} = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$$

Example: Find $\mathbf{a} \times \mathbf{b}$ and hence the area of the triangle with vertices A, B, C where $\overrightarrow{AB} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ 3 & 0 & 2 \end{vmatrix} = \mathbf{i}(4-0) - \mathbf{j}(2+3) + \mathbf{k}(0-6) = \begin{pmatrix} 4 \\ -5 \\ -6 \end{pmatrix}$$

$|\mathbf{a} \times \mathbf{b}| = \sqrt{16 + 25 + 36} = \sqrt{77}$
Area of triangle $= \frac{1}{2}\sqrt{77}$

4. Equation of a Line in 3D

A line through point A with position vector $\mathbf{a}$ and direction vector $\mathbf{b}$ has equation:

Vector form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, where $\lambda \in \mathbb{R}$ is a parameter.

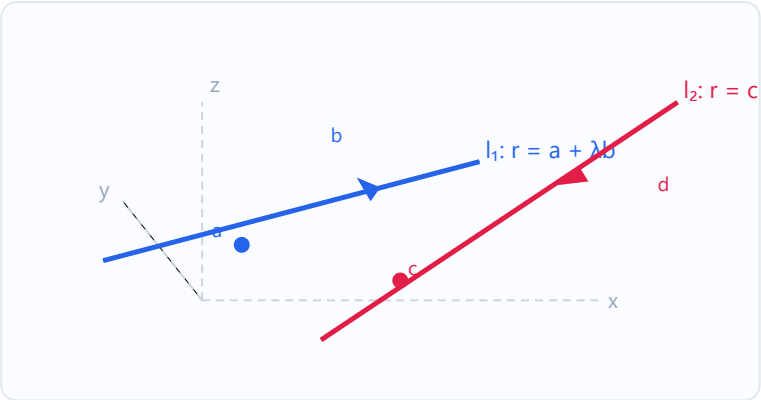
Parametric form:

$$x = a_1 + \lambda b_1, \quad y = a_2 + \lambda b_2, \quad z = a_3 + \lambda b_3$$

Cartesian (symmetric) form:

$$\frac{x-a_1}{b_1}=\frac{y-a_2}{b_2}=\frac{z-a_3}{b_3}=\lambda$$

(provided all components of $\mathbf{b}$ are non-zero)



Example: Write the line through $A=(2,-1,3)$ and $B=(5,1,-2)$ in all three forms.

Direction vector: $\overrightarrow{AB}=\begin{pmatrix}3\\2\\-5\end{pmatrix}$

Vector form: $\mathbf{r}=\begin{pmatrix}2\\-1\\3\end{pmatrix}+\lambda\begin{pmatrix}3\\2\\-5\end{pmatrix}$

Parametric: $x=2+3\lambda,\ y=-1+2\lambda,\ z=3-5\lambda$

Cartesian: $\dfrac{x-2}{3}=\dfrac{y+1}{2}=\dfrac{z-3}{-5}$

5. Equation of a Plane

A plane with normal vector $\mathbf{n}=\begin{pmatrix}a\\b\\c\end{pmatrix}$ passing through point A with position vector $\mathbf{a}$:

Scalar product form: $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} = d$

Cartesian form: $ax + by + cz = d$

Parametric form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ where $\mathbf{b}$ and $\mathbf{c}$ are two non-parallel direction vectors in the plane.

Example: Find the equation of the plane through $A(1,0,2)$, $B(3,1,0)$, $C(0,2,1)$.

Find two direction vectors: $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$

Normal: $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -2 \\ -1 & 2 & -1 \end{vmatrix}$

$= \mathbf{i}(-1+4) - \mathbf{j}(-2-2) + \mathbf{k}(4+1) = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$

$d = \mathbf{a} \cdot \mathbf{n} = 1(3) + 0(4) + 2(5) = 13$
Plane: $3x + 4y + 5z = 13$

6. Angles Between Lines and Planes

Angle between	Formula	Notes
Two lines (direction $\mathbf{b}_1, \mathbf{b}_2$)	$\cos\theta = \dfrac{\lvert\mathbf{b}_1 \cdot \mathbf{b}_2\rvert}{\lvert\mathbf{b}_1\rvert\lvert\mathbf{b}_2\rvert}$	Use $\lvert \cdot \rvert$ to get acute angle
Two planes (normals $\mathbf{n}_1, \mathbf{n}_2$)	$\cos\theta = \dfrac{\lvert\mathbf{n}_1 \cdot \mathbf{n}_2\rvert}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}$	Use $\lvert \cdot \rvert$ to get acute angle
Line (direction $\mathbf{b}$) and plane (normal $\mathbf{n}$)	$\sin\theta = \dfrac{\lvert\mathbf{b} \cdot \mathbf{n}\rvert}{\lvert\mathbf{b}\rvert\lvert\mathbf{n}\rvert}$	Note: $\sin$, not $\cos$

Exam tip: For the angle between a line and a plane, the formula uses $\sin$ (because the line makes an angle θ with the plane but $90^\circ - \theta$ with the normal). This is the most common error in this section.

7. Intersections

Line–Line Intersection (or Skew)

For two lines $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and $\mathbf{r} = \mathbf{c} + \mu\mathbf{d}$:

1. Set the x, y, z components equal to get three equations in λ and μ
2. Solve two of the equations for λ and μ
3. Check if these values satisfy the third equation
4. If yes: lines intersect (substitute to find point). If no: lines are skew or parallel

Parallel lines: Direction vectors are parallel ($\mathbf{b} = k\mathbf{d}$ for some scalar k).

Skew lines: Not parallel and do not intersect.

Line–Plane Intersection

Substitute the parametric equations of the line into the plane equation to find λ, then find the point.

Example: Find where $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ meets the plane $x - 2y + z = 5$.

Substitute parametric form: $(1 + 2\lambda) - 2(2 + \lambda) + (-1 + 3\lambda) = 5$
$1 + 2\lambda - 4 - 2\lambda - 1 + 3\lambda = 5$
$3\lambda - 4 = 5 \Rightarrow \lambda = 3$

Point: $x = 7, y = 5, z = 8$. Intersection at $(7, 5, 8)$.

Plane–Plane Intersection

Two non-parallel planes intersect in a line. Find the direction vector using the cross product of the two normals. Then find a specific point on the intersection (set one variable to 0 and solve the system of two

equations).

8. Distance Formulae

Distance from point P to plane $\mathbf{r} \cdot \mathbf{n} = d$:

$$\text{Distance} = \frac{|\mathbf{p} \cdot \mathbf{n} - d|}{|\mathbf{n}|}$$

where $\mathbf{p}$ is the position vector of P.

In Cartesian form (plane $ax + by + cz = d$, point (x_0, y_0, z_0)):

$$\text{Distance} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Distance from point A to line $\mathbf{r} = \mathbf{b} + \lambda\mathbf{d}$:

$$\text{Distance} = \frac{|\overrightarrow{BA} \times \mathbf{d}|}{|\mathbf{d}|}$$

where B is any point on the line.

Shortest distance between two skew lines $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and $\mathbf{r} = \mathbf{c} + \mu\mathbf{d}$:

$$\text{Distance} = \frac{|(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|}$$

Example: Find the shortest distance between the skew lines

$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

$$\mathbf{b} \times \mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 - 0 \cdot (-1) \\ 0 \cdot 1 - 2 \cdot 2 \\ 2 \cdot (-1) - 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -3 \end{pmatrix}$$

$$|\mathbf{b} \times \mathbf{d}| = \sqrt{4 + 16 + 9} = \sqrt{29}$$

$$\mathbf{c} - \mathbf{a} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

$$(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d}) = -2 - 8 - 12 = -22$$

Shortest distance: $\dfrac{|-22|}{\sqrt{29}} = \dfrac{22}{\sqrt{29}} = \dfrac{22\sqrt{29}}{29}$

9. Summary of Key Vector Formulae

Result	Formula
Dot product	$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \cos\theta$
Cross product magnitude	$\lvert\mathbf{a} \times \mathbf{b}\rvert = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta$
Area of triangle	$\frac{1}{2}\lvert\mathbf{a} \times \mathbf{b}\rvert$
Line equation	$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$
Plane equation	$\mathbf{r} \cdot \mathbf{n} = d$ or $ax + by + cz = d$
Angle between lines	$\cos\theta = \frac{\lvert\mathbf{b}_1 \cdot \mathbf{b}_2\rvert}{\lvert\mathbf{b}_1\rvert\lvert\mathbf{b}_2\rvert}$
Angle between planes	$\cos\theta = \frac{\lvert\mathbf{n}_1 \cdot \mathbf{n}_2\rvert}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}$
Angle: line and plane	$\sin\theta = \frac{\lvert\mathbf{b} \cdot \mathbf{n}\rvert}{\lvert\mathbf{b}\rvert\lvert\mathbf{n}\rvert}$
Point to plane distance	$\frac{\lvert\mathbf{p} \cdot \mathbf{n} - d\rvert}{\lvert\mathbf{n}\rvert}$

Point to line distance	$\frac{\lvert \overrightarrow{BA} \times \mathbf{d} \rvert}{\lvert \mathbf{d} \rvert}$
Skew lines distance	$\frac{\lvert (\mathbf{c}-\mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d}) \rvert}{\lvert \mathbf{b} \times \mathbf{d} \rvert}$

Common exam mistakes in vectors:

- **Line–plane angle:** Must use $\sin\theta$ with the NORMAL, not $\cos\theta$.
- **Skew vs parallel:** Always check both conditions. Skew means not parallel AND not intersecting.
- **Normal to plane from parametric form:** Take the cross product of the two direction vectors.
- **Shortest distance:** Always verify the lines are actually skew before applying the formula.
- **Exact answers:** Leave distances in surd form unless told to give a decimal.

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